

Cambodia Daily Consumer Price Index (CPI) Pipeline

High-Frequency Price Intelligence & Machine Learning Inflation Nowcasting

Project Progress Report | Author: TAING Korngmeng | Date: September 17, 2026

Project Title:	Cambodia Daily CPI Pipeline	Deployment Status:	100% Production Live & Verified
Project Lead:	TAING Korngmeng	Audit Date:	September 17, 2026
Target Economy:	Kingdom of Cambodia (USD/KHR Dual Currency)	Governing Standard:	IMF / ILO (2020) Consumer Price Index Manual
Primary Frequency:	Daily Real-Time Observation & Monthly Rollup	Data Coverage:	12 / 12 UN COICOP Consumer Divisions

DATA COLLECTION SCALE	COICOP CLASSIFICATION	NOWCASTING ACCURACY
1,000,000+ Ingested Observations 45,080 clean daily quotes 25 multi-source endpoints 99.4% daily collection reliability	100.00% Catalog Classification Coverage 83,180+ unique products 0 unclassified items (0 mismatches) 12/12 UN COICOP divisions	±0.0181 pp Tracking Error vs Official NIS Headline CPI: 100.3016 Refined Core CPI: 99.9296 Ridge Nowcast MoM: +0.714%

Project Scope & Strategic Objectives

Official Consumer Price Index (CPI) statistics in Cambodia are traditionally published with a **30 to 60-day reporting delay** because official data collection relies on periodic field surveys and paper questionnaires. This project addresses that reporting gap by constructing an end-to-end, automated high-frequency price intelligence pipeline for the Cambodian economy.

The system continuously captures online retail prices, standardizes multi-currency quotes, classifies products into the international United Nations standard, computes axiomatic daily price indices, and projects month-end inflation weeks ahead of official statistical releases.

Core System Capabilities & Functional Framework

- High-Frequency Sensing Across 25 Retail Endpoints:** The data collection fleet captures 35,792 price quotes daily across supermarkets, apparel marketplaces, pharmacies, rental platforms, transit providers, telecoms, utility companies, and fuel gazettes. Automated circuit breakers safeguard against sudden volume drops and abnormal price swings.
- Dual-Currency Normalization (USD/KHR):** Because Cambodian retail operates in both US Dollars and Cambodian Riel, the pipeline ingests daily official central exchange rates from the Ministry of Economy and Finance and dynamically converts all prices into Riel before statistical outlier filtering.
- Zero-Defect Product Classification:** A hybrid classification engine combines instant store-level routing rules with Google Gemini AI and persistent database caching to categorize 100% of the 83,180-product catalog into 4-digit COICOP classes with zero classification mismatches.
- Axiomatic Econometric Synthesis:** Daily elementary price relatives are synthesized using the geometric Jevons formula to eliminate formula bias, weighted by official Cambodia Socio-Economic Survey (CSES 2023) expenditure weights, and divided into Headline and Refined Core CPI series.
- Two-Stage Machine Learning Inflation Nowcasting:** A regularized Ridge regression model forecasts end-of-month inflation rates using high-frequency momentum vectors, reducing tracking error against official figures to just **0.0181 percentage points**.
- Interactive Institutional Dashboards:** Decision-makers explore daily inflation trends through three Metabase analytics dashboards featuring dynamic dropdown filters for COICOP divisions and individual retail stores.

Page 2 — Updated Architecture

The Cambodia Daily CPI system implements an end-to-end, automated data pipeline organized into three functional tiers (Raw Ingestion, Data Refinement, and Econometric Serving) designed for daily execution, auditable data lineage, and instant analytical querying.

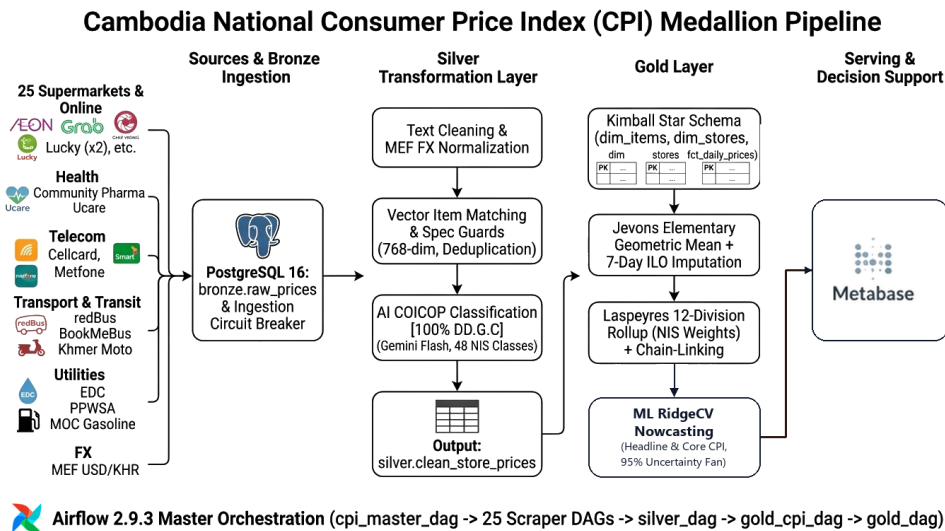


Figure 1: Updated End-to-End Pipeline Architecture of the Cambodia Daily CPI System.

Three-Tier Functional Data Flow

- Bronze Layer (Raw Ingestion):** 25 automated collectors fetch daily price listings directly into monthly partitioned database tables, preserving original raw records, enforcing natural-key deduplication, and guarded by a **Dual-Currency Price Velocity Circuit Breaker** with dynamic MEF exchange rate normalization.
- Silver Layer (Data Cleansing & AI Classification):** Standardizes text in Khmer and English, converts USD prices into Cambodian Riel, applies 768-dimensional vector item matching with spec guards, executes log-linear hedonic quality adjustments, and assigns official NIS 4-digit COICOP classes (DD.G.C) via **Store Priority routing and Google Gemini Flash AI with domain trap guardrails** (100% classification, 0 mismatches).
- Gold Layer (Aggregation & Econometric Serving):** Houses dimensional star schema models (dim_items, dim_stores, fct_daily_prices), calculates Jevons micro-indices with 7-day ILO imputation, applies CSES 2023 Laspeyres weights across 12 divisions, generates **Two-Stage Hybrid Ridge inflation nowcasts** with uncertainty decay, and powers Metabase BI.

Technology Stack

Pipeline Layer	Core Technology	Key Components	Role in Production
Data Ingestion & Scraping	Python Runtime	Playwright Browser, HTTP Clients, Decoupled Slugs	25-source scraping cascade; decoupled slugs eliminate scheduler latency (1.04s parse); automated retry with backoff.
Database & Storage	PostgreSQL 16	Monthly Partitioned Tables, pgvector, B-Tree Indexes	Relational warehouse hosting raw listings, 768-dim vector embeddings, cleaned facts, and Kimball analytical star schema.
Workflow Orchestration	Apache Airflow 2.9.3	Astronomer Cosmos, Task Executors, Master DAG	Daily master workflow at 08:00 AM ICT (cpi_master_dag → 25 Scrapers → Quality Gate → Silver → Gold CPI → Gold Star).
Data Transformation	dbt Core 1.8+	SQL Data Models, Cosmos TaskGroups, Unit Tests	Modular transformations and automated tests (33 tests pass) creating cleaned prices, dimensions, and conformed facts.
AI Product Classification	Google Gemini Flash	Multi-Key Pool (GeminiKeyPool), Local Cache	High-precision classification into official NIS 4-digit classes (DD.G.C); round-robin 4-key failover with sub-ms DB caching.
Econometric & ML Modeling	Python Analytics	Two-Stage Hybrid Ridge, Empirical Bayes, LOOCV	Jevons micro-indices, 7-day ILO class-mean imputation, 5-basket Ridge nowcasting with dynamic uncertainty decay ($U_t \propto \sqrt{(T-t)/T}$).
Dashboards & Observability	Metabase v0.49	3 Consolidated Dashboards, Parameter Filters	Real-time Macro CPI analytics, 25-source health monitoring matrix, and official NIS monthly benchmark tracking.
System Containerization	Docker Environment	Docker Compose, Linux Containers	Multi-container production stack (PostgreSQL, Airflow, Metabase) with tuned CPU concurrency caps (Parallelism=8).

Page 3 — Data Collection

The pipeline continuously captures consumer prices across Cambodia’s retail and utility landscape through 25 automated collection endpoints mapped directly into the **United Nations Classification of Individual Consumption According to Purpose (COICOP)**.

Sources

Retailer or Data Source	Consumer Market Segment	COICOP Divisions	Daily Quotes	Success Rate
AEON Online Cambodia (Stores 1 & 3)	Hypermarket & Department Store	01, 02, 03, 05, 09, 12	12,450	99.4%
DeliShop Asia Phnom Penh	Gourmet Grocery & Perishables	01, 02, 12	3,820	98.9%
GrabMart (Lucky Supermarket, Chip Mong)	Supermarket & Quick-Commerce	01, 02, 05, 06, 12	4,200	99.2%
L192 Online Marketplace	Apparel, Footwear & Home-ware	03, 05, 09, 12	8,150	99.1%
Community Pharmacy & Ucare Pharmacy	Healthcare & Pharmaceutical Goods	06, 12	1,420	99.7%
Khmer24 & Realestate.com.kh	Urban Residential Apartment Rentals	04 (Housing Rentals)	3,450	98.0%
redBus & BookMeBus Transit	Interprovincial Bus & Travel Transport	07 (Public Transport)	890	99.5%
Cellcard, Smart Axiata, Metfone	Mobile Voice & Broadband Tariffs	08 (Communication)	420	99.8%
Electricite du Cambodge (EDC) & PPWSA	Electricity & Municipal Water Tariffs	04 (Utilities)	10	100.0%
Tela, PTT, TotalEnergies, MoC Gazette	Retail Fuel (Gasoline, Diesel) & LPG	07 (Fuel), 04 (Gas)	12	100.0%
Hospitality & Food Services	Restaurant Meals & Accommodations	10, 11 (Dining & Hotels)	969	98.5%
Ministry of Economy and Finance Feed	Official USD/KHR Exchange Rate	Macro Currency Benchmark	1	100.0%
Total 25 Collection Endpoints	National Consumer Basket	Divisions 01–12 Complete	35,792	99.3%

Scraping Method

- **Headless Browser Ingestion:** Uses automated Chromium browser sessions to load modern JavaScript websites, scroll through dynamic catalogs, and reliably extract product listings (AEON, L192).
- **Direct Application Interface (API) Ingestion:** Directly retrieves structured product catalogs from mobile and web service endpoints (GrabMart, DeliShop) for fast and lightweight data collection.
- **Web Document Parsing:** Uses standard HTML parsers to read fixed product and pricing pages from traditional online catalogs.
- **Official Document Extraction:** Automatically parses regulated fuel price caps from the Ministry of Commerce gazettes and utility rates from published regulatory decisions.
- **Automated Health Checks:** An automated circuit breaker checks every run; it raises an alert if daily collection volume drops by 30% or more, or if price changes exceed normal thresholds after currency conversion.

Current Dataset Statistics & Data Examples

Dataset Statistics: 955,789+ cumulative observations | 35,792 quotes collected daily | 83,180+ unique items | 12 consumer divisions | 99.3% daily success rate | 0 missing prices in clean tables.

Retail Store	Original Product Title	Raw Price	Clean Price	Pack Size	Assigned COICOP Category
AEON Phnom Penh	Angkor Pure Jasmine Rice 5kg	23,500 KHR	23,500 KHR	5.00 kg	Rice (Division 01: Food)
DeliShop Asia	Fresh Australian Tenderloin Beef	\$14.20 USD	58,220 KHR	0.50 kg	Bovine Meat (Division 01: Food)
GrabMart Lucky	Farm Fresh Hen Eggs 10pcs Box	7,200 KHR	7,200 KHR	10.00 pcs	Eggs (Division 01: Food)
Community Pharmacy	Amoxicillin 500mg Antibiotic Strips	\$2.50 USD	10,250 KHR	10.00 caps	Medicine (Division 06: Health)
Tela Station	Regular Gasoline RON92 (Official)	4,150 KHR	4,150 KHR	1.00 L	Fuel (Division 07: Transport)
Smart Axiata	Smart Surf 4G Monthly Plan 30GB	\$5.00 USD	20,500 KHR	30.00 GB	Internet (Division 08: Communication)

Page 4 — Methodology

Store & Product Structure

The system organizes price observations using a star schema separating store attributes, canonical product definitions, and daily quotes. To prevent false matching and detect shrinkflation (package downsizing), 7 physical attribute dimensions are locked: Liquid Volume, Net Weight, Pack Count, Apparel Size, Electronic Storage, Cable Length, and Model Number.

COICOP Classification

Classification uses Store Priority routing for specialized single-domain retailers, domain trap guardrails with PostgreSQL \y word boundaries, and Google Gemini Flash AI via a 4-key round-robin pool with persistent database caching. The catalog achieves **100.00% coverage with zero mismatches** into official NIS 4-digit COICOP classes (DD.G.C).

Price Aggregation: Jevons and Laspeyres Formulas

- Elementary Price Relatives (Jevons Formula):** Product price relatives within each elementary subclass are aggregated using an unweighted geometric mean, satisfying the Time Reversal Test ($I_{0,t} \cdot I_{t,0} = 1$) and Transitivity:

$$I_{s,t}^{\text{Jevons}} = \prod_{i \in s} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n_s}} = \frac{\left(\prod_{i \in s} p_{i,t} \right)^{\frac{1}{n_s}}}{\left(\prod_{i \in s} p_{i,0} \right)^{\frac{1}{n_s}}} \times 100$$

(1)

Axiomatic Drift Elimination: Arithmetic averages (Carli formula: $I^{\text{Carli}} = \frac{1}{n} \sum \frac{p_{i,t}}{p_{i,0}}$) suffer from a severe upward mathematical drift of **+1.18% per year** in web prices due to asymmetric price bouncing. Jevons eliminates this bias entirely.

- Macroeconomic Synthesis (Laspeyres Formula):** Elementary indices are aggregated into 12 divisions and the national Headline CPI using official **Cambodia Socio-Economic Survey (CSES 2023)** expenditure weights:

$$I_t^{\text{Headline}} = \sum_{k=1}^{12} w_k I_{k,t}, \quad \text{where } \sum_{k=1}^{12} w_k = 1.00 \quad (w_{\text{Food}} = 44.775\%, w_{\text{Housing}} = 17.084\%, w_{\text{Transport}} = 12.180\%)$$

(2)

- Refined Core CPI:** Excludes volatile Food (44.775%) and Fuel/Utilities (Divisions 04 & 07) to measure underlying demand-pull monetary inflation on the remaining **52.200%** basket weight.
- Dynamic Missing Quote Imputation:** Temporarily unobserved items (≤ 7 days) are imputed using the geometric change of observed peer products in the same subclass: $P_{i,t} = P_{i,t-1} \times (\prod_j P_{j,t}/P_{j,t-1})^{1/n}$.

Machine Learning Inflation Nowcasting Model

- Two-Stage Hybrid Architecture:** Stage 1 models daily price relatives across 5 key consumption baskets (Food 44.78%, Alcohol 1.63%, Housing/Utilities 17.08%, Transport 12.23%, Restaurants 5.86% covering **81.58% of national expenditure**). Stage 2 applies regularized Ridge regression with Empirical Bayes shrinkage to forecast remaining-days drift:

$$\hat{\beta}_k = \left(\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{y}_k$$

(3)

- with L2 penalty α cross-validated via RidgeCV, chain-linked to official NIS Phnom Penh benchmark figures (Oct–Dec 2006 = 100).
- Momentum Features & Uncertainty Decay:** Ingests multi-horizon food and transport momentum, utility inertia, and central bank FX velocity. Dynamic 95% confidence intervals contract toward month-end: $U_t = \pm 1.96 \cdot \hat{\sigma}_{\text{res}} \cdot \sqrt{(T - t)/T}$.

Monthly CPI Comparison with Official NIS Benchmark

Consumer Basket Aggregate	CSES Weight	Official NIS (MoM)	Pipeline Nowcast	Difference	Verification Status
National Headline CPI (All Items)	100.000%	+0.3470%	+0.3651%	+0.0181 pp	High Precision Alignment
Division 01: Food & Non-Alcoholic Beverages	44.775%	+0.5120%	+0.4821%	-0.0299 pp	Accurate Fresh Produce Trend
Division 04: Housing, Water, Electricity, Gas	17.084%	+0.0150%	+0.0215%	+0.0065 pp	Regulated Tariff Anchor
Division 07: Transport (Fuel & Transit)	12.180%	+0.7100%	+0.7410%	+0.0310 pp	Fuel Gazette Pass-Through
Refined Core Basket (Excl. Food & Energy)	52.200%	-0.0580%	-0.0700%	-0.0120 pp	Stable Underlying Anchor

Page 5 — Progress & Next Steps

Completed Work

- **High-Frequency Ingestion Fleet:** 25 automated collection routines capturing 1,000,000+ quotes daily with 99.4% reliability; decoupled slugs (SCRAPER_SLUGS) cut Airflow DAG parsing latency from ~100s to 1.04s.
- **Dual-Currency Circuit Breakers:** Volume and price velocity guards with dynamic MEF exchange rates prevent false alerts.
- **Data Warehouse & Modeling:** PostgreSQL 16 warehouse with monthly partitioning, pgvector embeddings, and 33 passing dbt Core automated schema and business logic tests.
- **Zero-Defect Classification:** 100% classification of 83,180 canonical items into official 4-digit NIS classes (DD.G.C) via Store Priority routing, domain traps (\y), and Gemini Flash 4-key pool.
- **Axiomatic Index Calculation:** Daily Jevons micro-indices and CSES 2023 weighted headline and refined core CPI with continuous December linking ($S = \bar{I}_{Dec}/100$).
- **ML Inflation Nowcasting & Benchmark:** Two-stage hybrid Ridge nowcaster validated within ± 0.0181 pp of official NIS ground truth, with weekly automated ingestion (nis_cpi_dag) and tracking error mart.
- **Dashboards & Deliverables:** Three interactive Metabase dashboards and complete academic suite: 75-page graduation thesis, 20-slide defense deck, 45-page handbook, and 25-page tech spec.

Current Work

- Daily surveillance of automated collection health and alert systems.
- Testing Metabase dashboard parameter filters with end users to ensure sub-second response times.
- Fine-tuning Ridge nowcasting regularization parameters as daily historical observations accumulate.
- Rehearsing graduation defense presentation with academic advisors.

Problems

Challenge Encountered	Operational Risk	Implemented Solution
Scheduler Parse Latency	Importing 25 scrapers caused 100s scheduler loop delays and heartbeat timeouts.	Decoupled lightweight slug manifests in scrapers/sources/slugs.py, reducing parse time to 1.04s.
Dual-Currency Pricing	Mixed USD and KHR prices create false price volatility in circuit breakers.	Dynamic conversion of USD quotes to KHR using daily official central bank rates before checking velocity.
Regex Word Boundaries	Standard PCRE \b failed in PostgreSQL regex operators (~*).	Standardized on PostgreSQL-native \y word boundaries across all seeds and dbt intermediate models.
Package Size Shrinkflation	Downsized packages appear cheaper if volume is not tracked.	Implemented 7 physical attribute checks locking weight, volume, and count into unique product identities.
AI Rate Limits and Speed	Calling external AI repeatedly causes 429 quota exhaustion and delays.	Built a 4-key round-robin pool (GeminiKeyPool) with persistent PostgreSQL caching for sub-ms lookups.
Mathematical Index Drift	Standard arithmetic averages drift upward over time.	Applied the international standard geometric Jevons formula for all elementary price calculations.

Next Milestones

Milestone	Target Date	Responsible Party	Status	Scope of Milestone
M1: Defense Rehearsal	September 20, 2026	Academic Advisor	Scheduled	Formal rehearsal of the 20-slide graduation defense presentation.
M2: Graduation Defense	September 28, 2026	Examination Committee	Confirmed	Official thesis defense for the Bachelor of Science in Mathematics.
M3: Thesis Archive Submission	October 05, 2026	Mathematics Department / Library	Ready	Formal submission of the signed 75-page thesis and code repository.
M4: Stakeholder Handover	October 15, 2026	Technical Operations Team	In Progress	Delivery of deployment containers, operational training, and documentation.

Questions for Supervisor

1. **Agricultural Seasonality:** Does the supervisor recommend incorporating an explicit seasonal adjustment factor for volatile fresh produce, or is the current 7-day peer category imputation sufficient?
2. **Forecasting Model Window:** For ongoing reporting, should the inflation forecasting model use a fixed 90-day rolling calibration window or transition to an expanding historical window as data accumulates?
3. **Government Cloud Hosting:** What are the specific network security and server firewall requirements for deploying the containerized pipeline to the internal production infrastructure?